

Systems of Linear Equations



Setting up the augmented matrix

- 1 Write the augmented matrix for the system $2x + y - z = 3$, $x - 3y + 2z = -1$, $4x + y + z = 5$ (give it as its three rows).
 - 2 Write the augmented matrix for the system $x - y = 4$, $3x + 2y = 7$.
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Elementary row operations

- 3 List the three types of elementary row operations.
 - 4 Given the augmented matrix with rows $\langle 1, 2, 5 \rangle$, $\langle 3, -1, 1 \rangle$, apply the row operation $R_2 \rightarrow R_2 - 3R_1$ and give the new Row 2.
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Solving 2x2 systems by elimination

- 5 Solve $2x + 3y = 7$, $x - y = 1$.
 - 6 Solve $3x - 2y = 4$, $5x + y = 11$.
 - 7 Solve $4x + y = 10$, $2x - 3y = -8$.
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Solving 3x3 systems by Gaussian elimination

- 8 Solve $x + y + z = 6$, $2y + 5z = -4$, $2x + 5y - z = 27$.
 - 9 Solve $x + 2y + z = 4$, $y - z = 1$, $3x - y + 2z = 1$.
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Row echelon form and back-substitution

- 10 A system's augmented matrix has been reduced to row echelon form with rows $\langle 1, 2, -1, 4 \rangle$, $\langle 0, 1, 3, -2 \rangle$, $\langle 0, 0, 1, 5 \rangle$ (variables x, y, z). Use back-substitution to solve.
 - 11 Explain why row echelon form makes a system easy to solve, in one or two sentences.
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Inconsistent and dependent systems

- 12 Solve (or classify) the system $x + y = 3$, $2x + 2y = 8$.
- 13 Solve (or classify) the system $x + y = 3$, $2x + 2y = 6$.
- 14 For the system $x - 2y = 4$, $-3x + 6y = k$, find the value of k that makes the system have infinitely many solutions (rather than no solution).
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Why row operations preserve the solution set

- 15 Explain briefly why multiplying a row by a nonzero constant c doesn't change a system's solution set.
- 16 Why is the restriction $c \neq 0$ needed for this row operation?
- 17 Explain briefly why adding a multiple of one row to another row also preserves the solution set.
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Systems with a parameter

- 18 For what value(s) of k does the system $x + ky = 3$, $2x + 4y = 6$ have infinitely many solutions?
- 19 For what value of k does the system $2x - y = 5$, $4x + ky = 1$ have NO unique solution (i.e., the coefficient matrix is singular)?
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Application: mixture and rate problems

- 20 A chemist mixes a 10% acid solution with a 30% acid solution to make 20 L of a 15% solution. Set up the system of equations (let x , y be the liters of each solution used).
- 21 Solve the mixture system set up above.
- 22 Two trains leave stations 300 km apart, heading toward each other. One travels at x km/h, the other at y km/h, and they meet after 2 hours: $2x + 2y = 300$. If the faster train's speed is $y = x + 20$, set up and solve the system for x and y .
- 23 A theater sells adult tickets for 12 dollars and child tickets for 7 dollars. On a night with 200 tickets sold for a total of 1900 dollars, set up and solve the system for the number of adult (a) and child (c) tickets sold.

- 24** Explain why real-world word problems like the two above always translate into a LINEAR system (rather than, say, a quadratic one), given the quantities described.